Build Your Own Email Finding/Verification System for $5/Month

A self-hosted email verification pipeline that finds and validates business emails at **zero per-email cost** — running on three free tools and a $5/month VPS. No expensive third-party API subscriptions, no per-result billing, no rate limits imposed by someone else's infrastructure.

INFRASTRUCTURE GUIDE | ZERO PER-EMAIL COST | SELF-HOSTED

THE STACK

Three Free Tools Do All the Work

The system is built on three complementary components, each handling a different class of email verification problem. Together, they cover nearly every corporate domain scenario — from straightforward SMTP verification to locked-down Microsoft 365 environments.

SMTP Verifier

Custom Python service running on your VPS. Talks directly to mail servers using SMTP protocol to confirm whether an email address exists — no middleman, no cost per check.

Reacher

Open-source Docker container that handles edge cases your SMTP verifier can't. Runs alongside your Python service and is invoked automatically when needed.

M365 Identity Check

Free public Microsoft API — no authentication required. Bypasses corporate email gateways like Proofpoint, Mimecast, and Barracuda that block traditional SMTP verification.

CAPABILITIES

What the System Can Do

Scale & Speed

- Process approximately **1 contact per second** with 15 concurrent workers
- Classify thousands of domains in bulk: catch-all, responsive, gateway-blocked, or dead
- Short-circuit verification — stops the moment a valid email is found
- Assign best-guess emails for catch-all domains automatically

Reliability & Intelligence

- Fully **resumable with checkpoint files** — crashes don't lose progress
- Bypass corporate gateways (Proofpoint, Mimecast, Barracuda) via M365 identity checks
- Verify email patterns against real mail servers via SMTP
- Cache domain classifications — never re-probe a domain you've already classified

PREREQUISITES & COST

Total Cost: ~$5/Month + $2/Year

You need exactly four things to get started. Three are free or nearly free. One is a subscription you likely already have.

Item	Cost	Where to Get It
Hetzner CX23 VPS	$4.91/month	hetzner.com — Helsinki or Nuremberg location
Cheap Domain	~$2/year	Porkbun or Namecheap — used as SMTP sender identity
Cloudflare DNS	Free	cloudflare.com — DNS management and records
Claude Code	Existing subscription	claude.ai — writes all pipeline code for you

Why Hetzner? European data centers, port 25 available (on request), reliable uptime, and the lowest cost-to-performance ratio for this workload. Hetzner approves port 25 unblocking within 24 hours via support ticket.

ARCHITECTURE

The Verification Waterfall

Every contact flows through a cascading waterfall. Each step is only attempted if the previous one fails or is inapplicable. This design minimizes API calls, cuts costs, and maximizes speed. The system is self-terminating — once a valid email is found, it stops immediately.

Step Zero is the highest-leverage optimization in the entire system. By sending one fake email per domain upfront, it classifies entire domains before spending any real verification calls — saving over 80% of SMTP requests on typical contact lists.

PATTERN STRATEGY

The 8 Email Patterns, Tried in Order

SMTP verification tries these eight patterns sequentially, stopping the moment a valid hit is confirmed. The order is calibrated by real-world frequency across business email conventions — **first.last@domain covers 50% of cases alone**.

01

first.last@domain

Covers 50% of all business emails. Always try first.

02

flast@domain

20% frequency. Initial + last name, no separator.

03

first@domain

8% — common at smaller companies and startups.

04

firstlast@domain

6% — full name concatenated, no separator.

01

first_last@domain

5% — underscore separator variant.

02

last.first@domain

4% — reversed order, common in some European companies.

03

firstl@domain

3% — first name plus last initial.

04

f.last@domain

2% — first initial plus full last name with dot.

THE SECRET WEAPON

M365 Identity Check — Bypassing Every Gateway

Corporate email gateways like Proofpoint and Mimecast are designed to block SMTP verification probes. The M365 Identity Check sidesteps all of them entirely — using a **free, unauthenticated Microsoft public API** to confirm whether a user account exists in a tenant's directory.

1 Check if Domain Uses Microsoft 365

`GET https://login.microsoftonline.com/getuserrealm.srf?login=anyone@domain.com`

Look for `NameSpaceType = "Managed"` in the response. If present, the domain runs on M365 — proceed to Step 2.

2 Check if the Specific User Exists

`POST https://login.microsoftonline.com/common/GetCredentialType` with body `{"Username": "first.last@domain.com"}`

`IfExistsResult = 0` → user exists, email is valid. `IfExistsResult = 1` → not found, try next pattern.

Why this works: Microsoft's authentication API must resolve user identities to present the correct login screen. That lookup happens before any firewall or gateway policy applies — making it permanently available and impossible to block without breaking M365 login itself.

SETUP — STEPS 1 & 2

Buy Your VPS, Domain, and Configure DNS

Step 1 — VPS & Domain

Go to **hetzner.com**, create a CX23 server running Ubuntu 24.04 in Helsinki or Nuremberg. Record the IP address and root password.

Critical: Submit a Hetzner support ticket to unblock port 25. This is mandatory — without it, SMTP verification won't work. Approval typically arrives within 24 hours.

Buy any cheap domain from **Porkbun or Namecheap** (~$2/year). This domain becomes your SMTP sender identity — it doesn't need to be meaningful, just registered.

Step 2 — DNS Configuration

Add your domain to Cloudflare (free plan). Three records are required — mail servers check all of them before accepting SMTP connections:

1. **A Record:** `mail1.yourdomain.com` → VPS IP address
2. **TXT Record (SPF):** `v=spf1 ip4:YOUR_VPS_IP ~all`
3. **Reverse DNS (in Hetzner console):** VPS IP → `mail1.yourdomain.com`

All three are non-negotiable. Missing any one causes mail servers to reject or ignore your verification probes.

SETUP — STEPS 3 & 4

Deploy Reacher and the SMTP Verifier

Step 3 — Deploy Reacher via Docker

SSH into your VPS and install dependencies:

```
apt update && apt install -y docker.io python3-dnspython
```

Run the Reacher container:

```
docker run -d --name reacher --restart always \
  -e RCH__FROM_EMAIL=verify@yourdomain.com \
  -e RCH__HELLO_NAME=mail1.yourdomain.com \
  -e RCH__HTTP_HOST=0.0.0.0 \
  -p 8080:8080 reacherhq/backend:latest
```

Test with a POST to `localhost:8080/v0/check_email` with body `{"to_email":"test@gmail.com"}`.

Step 4 — Deploy SMTP Verifier

Upload `smtp_api.py` to `/root/smtp_api.py` on the VPS. Create a systemd service at `/etc/systemd/system/smtp-verifier.service` with `Restart=always` and `RestartSec=5`.

Enable and start the service:

```
systemctl daemon-reload
systemctl enable smtp-verifier
systemctl start smtp-verifier
ufw allow 8081/tcp
```

Verify it's running:

```
curl -s http://localhost:8081/
```

Expected response: `{"service": "smtp-verifier", "status": "ok"}`

SETUP — STEPS 5 & 6

Set Up Claude Code and Build Your Pipeline

Step 5 — Create the Claude Code Skill

Create a skill file at `~/.claude/commands/email-verify.md` on your local machine. This file encodes your entire system context:

- Health check commands for both services
- API endpoints and request/response formats
- Step Zero domain classification logic
- The full waterfall flow and decision rules
- All non-negotiable operating rules

When you invoke `/email-verify` in Claude Code, it loads this full context automatically.

Step 6 — Tell Claude Code to Build the Pipeline

Give Claude Code this exact prompt:

> "Build an async Python email verification pipeline using the /email-verify skill. My CSV has first_name, last_name, corporate_website, linkedin_url. Find and verify emails for contacts missing email. Write results back to CSV. Use Step Zero first, then the waterfall. Make it resumable."

Claude Code reads your skill and builds the complete script: async Python with aiohttp, Step Zero classification, full waterfall, checkpoint resumability, and LinkedIn URL matching. **You write zero code yourself.**

NON-NEGOTIABLE RULES

Six Rules You Cannot Break

These rules aren't suggestions. Violating any of them will cause silent failures, false negatives, or wasted credits that are extremely difficult to diagnose after the fact.

🚫 Never Run SMTP from Local

Always use the VPS. Port 25 is blocked on residential and office ISPs. Running locally causes 30–40% false failure rates that look like valid results.

✅ Always Run Step Zero First

One fake probe per domain before any real patterns. This single step saves 80%+ of all verification calls on typical B2B contact lists.

📧 Always Use Your Registered Domain

Send as your configured domain — the one with DNS, SPF, and rDNS set up. Using anything else will cause mail servers to reject your SMTP handshake.

⚡ Short-Circuit on First Valid Hit

Stop the moment you find a valid email. Continuing to check remaining patterns wastes calls and credits with zero benefit.

💾 Cache Domain Classifications

Never re-probe a domain you've already classified. Store results in your checkpoint file and reuse them across all contacts at that domain.

🔑 Use LinkedIn URL as Row Key

Never use CSV row index as a stable identifier. Rows shift when files are edited. LinkedIn URL or another stable ID ensures checkpoint resume works correctly.

OPERATIONS

Troubleshooting Reference

Most issues are recoverable in under two minutes. Here's the complete diagnostic and resolution reference for every known failure mode.

Symptom	Likely Cause	Resolution
SMTP Verifier not responding	Service crashed or was never started	`ssh in → systemctl restart smtp-verifier`
Reacher returning errors	Docker container stopped	`ssh in → docker restart reacher`
VPS completely unreachable	Host-level issue or IP block	Check Hetzner console → redeploy server if needed
Gateway rate above 60%	Normal for enterprise B2B lists	Expected behavior — M365 check handles most of these
Low overall hit rate	DNS/SPF/rDNS misconfiguration or port 25 blocked	Verify all three DNS records and confirm port 25 unblocked with Hetzner

Pro tip: Run `systemctl status smtp-verifier` and `docker ps` as your first diagnostic step for any verification failure. 90% of issues are resolved by those two commands.

SUMMARY

What You've Built

A production-grade email verification infrastructure at a fraction of the cost of any SaaS alternative. Here's what's running when you're done:

$5

Monthly Cost

Total infrastructure spend. No per-email fees, no seat licenses, no surprise bills.

~1/s

Throughput

One contact verified per second with 15 concurrent async workers running in parallel.

80%+

Call Savings

Reduction in SMTP calls from Step Zero domain pre-classification alone.

3

Verification Layers

SMTP, M365 Identity Check, and optional LeadMagic — covering every domain type.

The system handles catch-all domains, gateway-blocked corporates, and dead domains automatically. Checkpointing ensures you never lose progress. Claude Code writes all the pipeline logic — you just configure and run.

Start Building Today

The entire system can be operational in an afternoon. Follow the six steps in order, respect the non-negotiable rules, and you'll have a production email verification pipeline running at near-zero cost before end of day.

Buy VPS & Domain

Hetzner CX23 + cheap domain. Submit port 25 ticket immediately.

Configure DNS

A record, SPF TXT, and reverse DNS. All three required.

Deploy Services

Reacher via Docker, SMTP Verifier via systemd. Both persistent.

Ship with Claude

Create skill file, run one prompt, get a complete async pipeline.

First run checklist: Port 25 unblocked ✓ · All three DNS records live ✓ · Both services returning 200 ✓ · Step Zero enabled in pipeline ✓ · LinkedIn URL set as checkpoint key ✓

If you need any help, please reach out.

LinkedIn: https://www.linkedin.com/in/go-to-market-engineer/

Email: nilesh@prospectspulse.com